

matreex: Simulating European forest dynamics with IPM.

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Summary

Integral projection models (IPMs) are powerful tools for studying the temporal dynamics of populations structured by continuous traits, allowing for predictions of changes in trait distributions over time ([Ellner et al., 2016](#)). Unlike individual-based or cohort-based models, which represent populations as discrete populations, IPMs describe populations as continuous populations, integrating over the uncertainty of demographic processes. This removes demographic stochasticity and results in fully deterministic simulations, which are complementary to individual-based models (IBMs).

Here, we introduce matreex, an R package specifically designed to build IPMs for European forest tree species. Our package includes pre-fitted species-specific growth, survival and recruitment functions that account for the effect of climate and competition, and functions to efficiently integrate IPMs and run temporal simulations of single-species or multispecies forest communities until equilibrium. This package complements existing R packages for IPMs, such as *ipmr* ([Metcalf et al., 2013](#)) and *IPMpack* ([Levin et al., 2021](#)), which are not specifically designed for forest ecosystems.

Statement of need

IPMs are rarely applied to forest ecosystems due to the complexity of tree growth kernels, which are challenging to integrate, making the construction of forest IPMs particularly difficult. matreex is an R package specifically designed to build IPMs and run simulations for European forests. These IPMs can be used by the forest ecology research community, to study population dynamic and forest communities trajectories. In matreex IPM simulations, it is also possible to include temporally variable climatic conditions, natural disturbances, harvesting scenarios and regional dispersal affecting population dynaminc depending on tree species sensitivity and stand structure.

State of the field

Other, more generalist, packages exist to build IPMs for various types of organisms (*IPMpack* ([Metcalf et al., 2013](#)), *ipmr* ([Levin et al., 2021](#))). However they focus on analysis on the matrix itself whereas matreex use the integrated matrices to simulate population dynamics. In addition matreex integrates numerous functions to run simulations for multispecies communities under various harvesting and disturbance scenarios.

39 Software design

40 A key feature of the matreex package is the development of IPM integration functions designed
41 for complex forest tree growth kernels. As trees have a small annual growth rate, most of
42 the integration effort is concentrated near the diagonal of the matrix. We achieve this by
43 combining different integration methods (Gauss-Legendre and Mid-bin) at different distances
44 of the diagonal, which helps speed up the integration. This speed is crucial as we integrate a
45 matrix per competition level (based on species basal area) to account for density dependence.
46 More details about the integration are provided in the [Building IPM vignette](#).

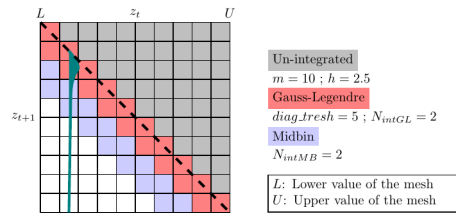


Figure 1: Figure 1: Combination of different integration methods. Dashed line is the identity where $z_t = z_t + 1$ and dark blue distribution is an expected distribution of the growth kernel.

47 A crucial development objective was to simplify the workflow for ecological researchers, who
48 may work on multispecies models with climate variation, disturbances, and harvesting. To
49 facilitate this, matreex provides fitted vital models for European tree species directly in the
50 package ([Barrere et al., 2024](#); [Guyennon et al., 2023](#); [Kunstler et al., 2021](#)), although other
51 new vital models can be used. The object-oriented architecture limits code complexity, allowing
52 users to focus on designing large simulation experiments to tackle their ecological questions.

```
53 library(matreex)
54 library(dplyr)
55 library(ggplot2)
56
57 # select a climate to run in
58 data("climate_species")
59 climate <- subset(
60   climate_species,
61   climate_label == "optimum" & sp == "Fagus_sylvatica",
62   select = -c(sp, climate_label)
63 )
64
65 # integrate ipms and store them in species object
66 Picea <- species(IPM = make_IPM(
67   species = "Picea_abies", fit = fit_Picea_abies,
68   climate = climate, clim_lab = "optimum clim",
69   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Picea_abies) * 1.1),
70   BA = 0:60
71 ), init_pop = def_initBA(1))
72 Betula <- species(IPM = make_IPM(
73   species = "Betula", fit = fit_Betula,
74   climate = climate, clim_lab = "optimum clim",
75   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Betula) * 1.1),
76   BA = 0:60
77 ), init_pop = def_initBA(1))
78 Fagus <- species(IPM = make_IPM(
79   species = "Fagus_sylvatica", fit = fit_Fagus_sylvatica,
```

```

80   climate = climate, clim_lab = "optimum clim",
81   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Fagus_sylvatica) * 1.1),
82   BA = 0:60
83 ), init_pop = def_initBA(1))
84 Abies <- species(IPM = make_IPM(
85   species = "Abies_alba", fit = fit_Abies_alba,
86   climate = climate, clim_lab = "optimum clim",
87   mesh = c(m = 700, L = 90, U = get_maxdbh(fit_Abies_alba) * 1.1),
88   BA = 0:60
89 ), init_pop = def_initBA(1))
90
91 # assemble species in a forest object
92 Forest <- forest(species = list(Picea = Picea, Abies = Abies,
93                                Fagus = Fagus, Betula = Betula))
94 set.seed(20260825) # The seed is here for initial population random functions.
95
96 # Run simulation and plot it
97 Sim <- sim_deter_forest(
98   Forest,
99   tlim = 1000,
100   equil_time = 1000, equil_dist = 50, equil_diff = 1,
101   SurfEch = 0.03,
102   verbose = TRUE
103 )
104 Sim %>%
105   dplyr::filter(var == "BAsp", ! equil) |>
106   ggplot(aes(x = time, y = value, color = species)) +
107   geom_line(linewidth = .4) +
108   ylab("Basal Area (m2)") + xlab("Simulation time (years)")

```

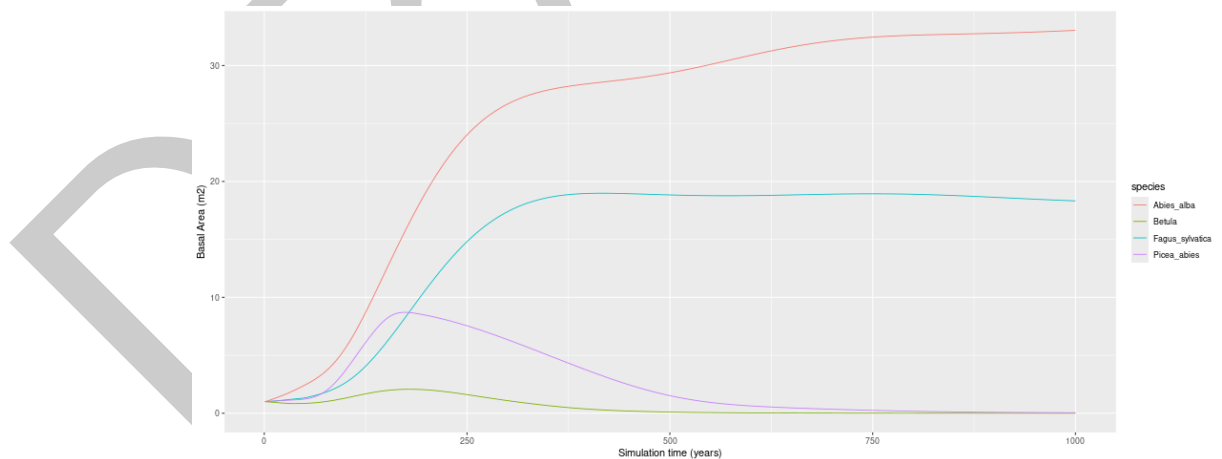


Figure 2: Figure 2: Simulation output for 4 species.

109 Climatic temporal variability

110 Modelling forest dynamics under fluctuating climatic conditions can be computationally
 111 expensive because the IPMs growth kernel must be integrated for every climatic condition. To
 112 avoid this high computation cost, we implement a new method involving the pre-integration
 113 of IPM growth matrix blocks for different mean growth rates. The IPM for each climatic
 114 condition is then reassembled from these mean growth rate IPM matrix blocks (mu matrix).

This allows to speed-up the simulations. This is described in the [matreex climate variation vignette](#).

Disturbance

One key originality of matreex is the possibility to apply storm, fire, biotic or snow disturbances of varying intensity (ranging from 0 to 1) at any time of the IPM simulations. When a disturbance strikes the stand a given year of the simulation, the survival function is replaced by the species-specific equations from Barrere et al. (2024). These equations quantify the annual mortality probability of a tree in a disturbed stand as a function of its species, diameter at breast height, stand structure, nature and intensity of the disturbance. A disturbance striking two different stands with the same intensity will thus result in different mortality rates, depending notably on the sensitivity of the tree species present in the stand. Disturbances in matreex are described in details in Barrere et al. (2024) and in Barrere et al. (in prep). Figure 3 shows an example of multispecies simulations with storm disturbance.

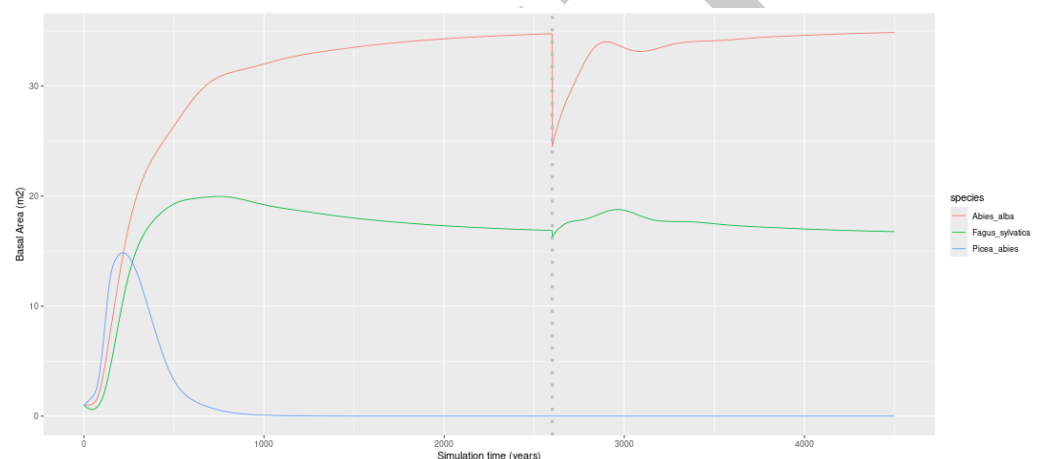


Figure 3: Simulation output for 3 species with a disturbance at $time = 2600$.

Regional dispersal

Most stand-scale forest dynamics models simulate closed systems, where only the tree species already present in the stand contribute to the recruitment of new trees. This limitation precludes the simulation of immigration from external species, which is a key process of forest dynamics, particularly under climate change. To overcome this limitation, we included the possibility to split the recruitment function in two components: (i) within-plot dispersal that depends on the sum of basal area of fecund tree species in the plot, and (ii) external dispersal that depends on the regional pool. This regional pool approach is extensively presented in Barrere et al. (in prep).

Harvesting

Since most temperate forests are managed, it is crucial to incorporate silvicultural interventions into the simulations. We implemented three management strategies:

- First, we implemented a simple constant annual harvesting rates accounting for the effect of the harvesting rates observed in NFI data used for the model calibration (Kunstler et al., 2021).
- Second, we implement an even-aged management. The objective is to apply typical even-aged harvesting, based on a single cohort. Trees are harvested with successive thinning during stand development until the final harvest. Thinning harvest are based

on the distance to a self-thinning boundary, based on Aussenac et al. (2021). This is easily connected with management guidelines.

- Third, we implement an uneven-aged harvesting. The uneven-aged harvest scenario consists in selective harvesting across all size classes with the objective to reach a stable size structure with continuous replacement of large mature trees. This scenario depends on the basal area of the stand and the size distribution of the trees (building on Lafond et al. (2014)). These three managements are described in the [matreex harvesting vignette](#).

Research impact statement

matreex was designed to be easily adapted to various research ideas and is in continuous development. It has already been used in several scientific publications tackling diverse scientific questions (Baranger et al., in prep; Barrere et al., 2024, in prep; Guyennon et al., 2023; Kunstler et al., 2021). The ability to simulate forests easily with a dedicated R package will help ecologists to analyse the effect of climate change, shifting disturbance regimes, and their interplay with forest management across European forests.

matreex is an open-source package made available under the MIT license. Installation and usage instructions can be found at the [website](#)

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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